A COMPUTER-FREE PROOF THAT A CONWAY 99-GRAPH HAS NO AUTOMORPHISM OF ORDER SEVEN

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Abstract. A *Conway 99-graph* is a strongly regular graph with parameters $(99, 14, 1, 2)$; whether one exists is a well-known open problem. We prove, without computer assistance, that such a graph has no automorphism of order 7. Consequently 7 does not divide the order of its automorphism group. This exclusion was previously obtained only with computer assistance, by the orbit-matrix computations of Behbahani and Lam. It also completes the computer-free treatment of the prime 7 begun by Cesarz and Woldar, who showed that if 7 divides the order of the automorphism group, then the group is cyclic of order 7.

Using the orbit structure and the double-counting identities of Cesarz and Woldar, an automorphism of order 7 yields a symmetric nonnegative integral 12×12 matrix C with $C\mathbf{1} = 12\mathbf{1}$, $Cu = 0$ and $C^2 + C = 12I + 12J - 2uu^{\mathsf{T}}$, where $u = (1^3, (-1)^3, 0^6)$. A rational-representation argument shows that C has zero diagonal. Then $G = 3C + 12I - 4J - 2uu^{\mathsf{T}}$ is 21 times an orthogonal projection of rank 4, so G is the Gram matrix of twelve vectors in $\mathbb{R}^4$. A short case analysis, done entirely by hand, shows that no such vectors exist. The theorem has also been verified formally in Lean 4 with Mathlib, starting from Mathlib's definition of a strongly regular graph.

1. Introduction

A strongly regular graph with parameters $(99, 14, 1, 2)$ is a graph on 99 vertices in which every edge lies in exactly one triangle and every non-edge is a diagonal of exactly one quadrilateral. Whether such a graph exists is a well-known open problem. Its possible existence was raised by Biggs [Bi71]. Conway offered a prize for its resolution and listed it as the second of his five \$1,000 problems [Co]. Following [CW25], we call such a graph a *Conway 99-graph*. Background on strongly regular graphs can be found in [BvM22].

One standard line of attack is to restrict the automorphism group $\operatorname{Aut}\Gamma$ of a putative Conway 99-graph Γ. Wilbrink [W84] studied the local structure of Γ. Makhnev and Minakova [MM04] used character-theoretic methods to show that $|\operatorname{Aut}\Gamma|$ divides $2 \cdot 3^3 \cdot 7 \cdot 11$, and that an automorphism of order 7 fixes exactly one vertex (see also [B09, Theorem 1.6]). Behbahani and Lam [B09, BL11] introduced orbit matrices for strongly regular graphs with a prescribed automorphism. With computer assistance they showed that every automorphism of prime order has order 2 or 3; in particular they excluded the prime 7. Crnković and Maksimović [CM20] ruled out automorphism groups of orders 6 and 9. In a 2014 MIT PRIMES report, Lou and Murin [LM14] analysed an automorphism of order 7 by hand. They showed that it has a unique fixed vertex and that its 15×15 orbit matrix has trace 0 or 14, but they did not exclude it.

Cesarz and Woldar [CW25] gave the first computer-free results of this kind. They proved that if 2 divides $|\operatorname{Aut}\Gamma|$ then $|\operatorname{Aut}\Gamma|$ divides 6, and that if 7 divides $|\operatorname{Aut}\Gamma|$ then $\operatorname{Aut}\Gamma \cong \mathbb{Z}_7$. To prove the second statement they reduced to the two groups $\mathbb{Z}_7$ and $\mathrm{Frob}(21)$ and then eliminated $\mathrm{Frob}(21)$. The cyclic case $\operatorname{Aut}\Gamma \cong \mathbb{Z}_7$ has therefore remained open from the computer-free point of view. SAT and constraint-programming approaches to the existence problem have been studied recently [K26, TS26]. Thakkar and Severini [TS26] describe the existence of a

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$\mathbb{Z}_7$-symmetric Conway 99-graph as open, and report that their prescribed-automorphism CP-SAT model stayed undecided after a 48-hour run. Two unrefereed computational treatments of the order-7 case were made public in 2026: an exhaustive orbit-matrix enumeration by Walker [Wa26], labelled by its author as awaiting independent reproduction, and a SAT refutation of a CNF encoding with a DRAT certificate by Machuca [Ma26].

In this paper we settle the order-7 case without computer assistance.

Theorem 1.1. *A strongly regular graph with parameters* $(99, 14, 1, 2)$ *has no automorphism of order* 7*. Consequently,* 7 *does not divide the order of its automorphism group.*

The second sentence follows from the first by Cauchy's theorem. The proof uses a single automorphism of order 7 and nothing about the rest of $\operatorname{Aut}\Gamma$. In particular it does not depend on the group-theoretic reduction of [CW25]. Combined with the results quoted above it gives the following computer-free statement.

Corollary 1.2. *Let* Γ *be a Conway* 99*-graph. Then either* $|\operatorname{Aut}\Gamma|$ *divides* 6*, or* $|\operatorname{Aut}\Gamma|$ *is odd and divides* $3^3 \cdot 11$*.*

Proof. By [MM04, Theorem 2.7], as recalled in [CW25], $|\operatorname{Aut}\Gamma|$ divides $2 \cdot 3^3 \cdot 7 \cdot 11$. By Theorem 1.1, $7 \nmid |\operatorname{Aut}\Gamma|$. If $|\operatorname{Aut}\Gamma|$ is even, it divides 6 by [CW25, Corollary 3.13]. □

Relation to previous work. The proof has two parts. The first part reduces an automorphism of order 7 to a problem about a 12×12 matrix; it is due mostly to Cesarz and Woldar [CW25] and, for some steps, to Lou and Murin [LM14]. The second part solves this matrix problem; to our knowledge, it does not appear in the literature. We adopt the labelling of [CW25, §2] and include proofs of the results we use, so that the paper is self-contained. The table below indicates the source of each step.

Step	Source
Local structure (Lemma 2.1)	standard; see [W84], [CW25, §2]
Unique fixed vertex, two 7-cycles in the neighbourhood (Lemmas 3.1 and 3.2)	[MM04]; [LM14, §8]; [CW25, Lemma 2.2, Remark 2.3]
Twelve exterior orbits and the identities $C\mathbf{1} = 12\mathbf{1}$, $Cu = 0$, $C^2 + C = 12I + 12J - 2uu^{\mathsf{T}}$ (Proposition 4.1)	orbit matrices [B09, BL11]; $Cu = 0$ is the L/R-balance of [CW25, Lemma 4.7]; the quadratic identity is [CW25, Eq. (5)], written in closed form
Trace of the orbit matrix in terms of eigenvalue multiplicities	[LM14, §8]; [CW25, proof of Theorem 3.11]
Zero diagonal (Proposition 5.2)	the divisibility $54 \equiv d \pmod 6$ is an instance of the character method for automorphisms of strongly regular graphs (cf. [MM04, BL11]); the same restriction, and the zero diagonal, appear in the computational report [Wa26]. Self-contained proof given here
Gram compression to $\mathbb{R}^4$ (Section 6), the LL/RR block (Section 7), the mixed orbits and the final contradiction (Section 8)	this paper

After the zero diagonal is established, the argument uses only elementary Euclidean geometry in $\mathbb{R}^4$ and the solution of a few small Diophantine equations. Every finite verification is written out in the text or in the appendices, so that it can be checked by hand. Independent machine checks and a complete formal verification in Lean are described in Section 9; they are not part of the proof.

Organization. Section 2 recalls the local structure of Γ. Section 3 describes the orbits of an automorphism of order 7, and Section 4 derives the identities satisfied by the exterior quotient matrix C. Section 5 shows that C has zero diagonal. Section 6 introduces the rank-4 Gram vectors. Section 7 classifies the 6×6 block of C on the LL and RR orbits, and Section 8 excludes each of the six resulting configurations.

2. Preliminaries

Let $\Gamma = (V, E)$ be a Conway 99-graph with adjacency matrix A. Write I for the identity matrix, J for the all-ones matrix and $\mathbf{1}$ for the all-ones vector. Counting common neighbours gives $A^2 = 14I + A + 2(J - I - A)$, that is,

$$A^2 + A = 12I + 2J. \tag{1}$$

Since $A\mathbf{1} = 14\mathbf{1}$, (1) shows that A acts on $\mathbf{1}^\perp$ with eigenvalues among the roots 3 and -4 of $X^2 + X - 12$. If f and g are their multiplicities, then $f + g = 98$ and $\operatorname{tr} A = 14 + 3f - 4g = 0$. So the spectrum of A is $14^1\, 3^{54}\, (-4)^{44}$.

For a vertex x write $\Gamma(x)$ for its set of neighbours and $\Gamma_2(x) = V \setminus (\{x\} \cup \Gamma(x))$, so that $|\Gamma_2(x)| = 84$.

Lemma 2.1 (local structure)**.** *Let $x \in V$.*

(a) $\Gamma(x)$ induces a perfect matching. For $a \in \Gamma(x)$ we write a' for its partner.
(b) Every $y \in \Gamma_2(x)$ has exactly two neighbours in $\Gamma(x)$, and they are non-adjacent.
(c) The map $y \mapsto \Gamma(y) \cap \Gamma(x)$ is a bijection from $\Gamma_2(x)$ onto the set of 84 non-adjacent pairs of vertices of $\Gamma(x)$.
(d) Every vertex of $\Gamma(x)$ has exactly 12 neighbours in $\Gamma_2(x)$.

Proof. (a) Since $\lambda = 1$, every $a \in \Gamma(x)$ has exactly one neighbour in $\Gamma(x)$. (b) Since $\mu = 2$, y has exactly two neighbours a, b in $\Gamma(x)$. If $a \sim b$, the adjacent vertices a, b would have the two common neighbours x, y, contradicting $\lambda = 1$. (c) Let $a, b \in \Gamma(x)$ be non-adjacent. They have two common neighbours, and x is one of them. The other one, z, is not in $\Gamma(x)$: otherwise $z = a'$ and $z = b'$, so $a = b$. Hence $z \in \Gamma_2(x)$, and z is the only vertex of $\Gamma_2(x)$ adjacent to both a and b. Together with (b) this gives the bijection. There are $\binom{14}{2} - 7 = 84$ non-adjacent pairs. (d) A vertex $a \in \Gamma(x)$ has 14 neighbours: x, a', and 12 vertices in $\Gamma_2(x)$. □

3. The orbits of an automorphism of order seven

From now on, $s \in \operatorname{Aut}\Gamma$ has order 7. Every $\langle s\rangle$-orbit on V has size 1 or 7. The following two lemmas are [CW25, Lemma 2.2 and Remark 2.3]; see also [MM04] and [LM14, §8].

Lemma 3.1. *s fixes exactly one vertex x. It fixes no vertex of $\Gamma(x)$ and no vertex of $\Gamma_2(x)$.*

Proof. The number of fixed vertices is congruent to $99 \equiv 1 \pmod 7$, so some vertex x is fixed. Then s preserves $\Gamma(x)$ and its matching, so the set N_0 of fixed vertices in $\Gamma(x)$ is closed under $a \mapsto a'$. In particular $|N_0|$ is even. Also $|N_0| \equiv |\Gamma(x)| = 14 \equiv 0 \pmod 7$, so $|N_0| \in \{0, 14\}$. If $|N_0| = 14$, then every $y \in \Gamma_2(x)$ is fixed, because by Lemma 2.1(c) it is determined by its two neighbours in $\Gamma(x)$. Then s fixes every vertex, which is impossible. Hence $N_0 = \emptyset$.

If $y \in \Gamma_2(x)$ were fixed, s would preserve the two-element set $\Gamma(y) \cap \Gamma(x)$. Since s has odd order it would fix both of its points, contradicting $N_0 = \emptyset$. Every vertex other than x lies in $\Gamma(x) \cup \Gamma_2(x)$, so x is the only fixed vertex. □

Lemma 3.2. *The vertices of $\Gamma(x)$ can be labelled ℓ_i, r_i $(i \in \mathbb{Z}/7)$ so that $\ell_i \sim r_i$, $s(\ell_i) = \ell_{i+1}$ and $s(r_i) = r_{i+1}$.*

Proof. The automorphism s permutes the seven edges of the matching on $\Gamma(x)$. If it fixed an edge $\{a, a'\}$, then s^2 would fix a, and so would $s = (s^2)^4$; this contradicts Lemma 3.1. So the seven edges form one $\langle s\rangle$-orbit. Choose $\ell_0 \in \Gamma(x)$ and put $\ell_i = s^i(\ell_0)$ and $r_i = \ell_i'$. Then $s(r_i) = s(\ell_i)' = r_{i+1}$. The edges $\{\ell_i, r_i\} = s^i\{\ell_0, r_0\}$ are the seven distinct matching edges, so the 14 labels are distinct. □

In the notation of [CW25], ℓ_i and r_i are the vertices iL and iR. Write $L = \{\ell_i\}$ and $R = \{r_i\}$. These are independent sets, and every vertex of R has exactly one neighbour in L.

By Lemma 2.1(c), each $y \in \Gamma_2(x)$ is labelled by a non-adjacent pair of $\Gamma(x)$, and s acts on labels by $i \mapsto i + 1$. The non-adjacent pairs are the 21 pairs $\{\ell_i, \ell_j\}$, the 21 pairs $\{r_i, r_j\}$, and the 42 pairs $\{\ell_i, r_j\}$ with $i \neq j$. Since 7 is odd, $\{\ell_i, \ell_{i+\delta}\}$ $(i \in \mathbb{Z}/7)$ are seven distinct pairs for each $\delta \neq 0$. Hence $\Gamma_2(x)$ splits into twelve orbits of size 7, which we call the *exterior orbits*:

- three *LL-orbits*, with labels $\{\ell_i, \ell_{i+\delta}\}$ for $\delta = 1, 2, 3$; their set is denoted $\mathcal{A}$;
- three *RR-orbits*, with labels $\{r_i, r_{i+\delta}\}$ for $\delta = 1, 2, 3$; their set is denoted $\mathcal{B}$;
- six *mixed orbits*, with labels $\{\ell_i, r_{i+\delta}\}$ for $\delta = 1, \dots, 6$; their set is denoted $\mathcal{M}$.

In [CW25, §4], where a Frobenius group of order 21 is assumed, these orbits are called $(LL)_i$, $(RR)_i$, $(LR)_i$ and $(RL)_i$ $(i = 1, 2, 3)$; without that assumption the six mixed orbits are simply indexed by δ. For an exterior orbit O put $u_O = 1, -1, 0$ according as $O \in \mathcal{A}$, $O \in \mathcal{B}$ or $O \in \mathcal{M}$. A vertex of O then has $1 + u_O$ neighbours in L and $1 - u_O$ neighbours in R.

4. The exterior quotient matrix

For exterior orbits O, O', let $C_{OO'}$ be the number of neighbours in O' of a vertex $y \in O$. This number does not depend on the choice of $y \in O$, since s is an automorphism. We order the orbits as $\mathcal{A}, \mathcal{B}, \mathcal{M}$, so that

$$u = (1, 1, 1, -1, -1, -1, 0, 0, 0, 0, 0, 0)^{\mathsf{T}}.$$

The matrix C is the restriction to $\Gamma_2(x)$ of the orbit matrix of $\langle s \rangle$ in the sense of [B09, BL11], and the identities below are the ones derived by double counting in [CW25, §4].

Proposition 4.1. *C is a symmetric 12×12 matrix with nonnegative integer entries, and*

$$C\mathbf{1} = 12\,\mathbf{1}, \tag{2}$$

$$Cu = 0, \tag{3}$$

$$C^2 + C = 12I + 12J - 2uu^{\mathsf{T}}. \tag{4}$$

Proof. The matrix is symmetric because $7C_{OO'}$ and $7C_{O'O}$ both count the edges between O and O' when $O \neq O'$. A vertex $y \in \Gamma_2(x)$ has 14 neighbours, two of them in $\Gamma(x)$ by Lemma 2.1(b), and it is not adjacent to x. This gives (2). A vertex of L has exactly $1 + u_{O'}$ neighbours in O', since both numbers are $1/7$ of the number of edges between L and O'. Similarly, a vertex of R has $1 - u_{O'}$ neighbours in O'.

Fix $y \in O$ and a set Z of vertices, and count the walks $y \to w \to z$ of length two with $z \in Z$. Grouping by z gives $\sum_{z \in Z} |\Gamma(y) \cap \Gamma(z)|$, and $|\Gamma(y) \cap \Gamma(z)|$ equals 14, 1 or 2 according as $z = y$, $z \sim y$, or neither. Grouping by w gives $\sum_{w \sim y} |\Gamma(w) \cap Z|$.

Take $Z = O'$. Write $\delta = 1$ if $O = O'$ and $\delta = 0$ otherwise. The first count is $14\delta + C_{OO'} + 2(7 - \delta - C_{OO'}) = 12\delta + 14 - C_{OO'}$. In the second count the neighbours of y in L, in R and in the exterior orbits contribute

$$(1 + u_O)(1 + u_{O'}) + (1 - u_O)(1 - u_{O'}) + \sum_{O''} C_{OO''}C_{O''O'} = 2 + 2u_O u_{O'} + (C^2)_{OO'}.$$

Equating the two counts gives (4).

Take $Z = L$. The vertex y has $1 + u_O$ neighbours in L, so the first count is $(1 + u_O) + 2(7 - 1 - u_O) = 13 - u_O$. In the second count, neighbours of y in L contribute 0 (as L is independent), neighbours in R contribute 1 each, and a neighbour in an exterior orbit O'' contributes $1 + u_{O''}$. So the second count is $(1 - u_O) + \sum_{O''} C_{OO''}(1 + u_{O''}) = 13 - u_O + (Cu)_O$, by (2). Hence $(Cu)_O = 0$. □

Taking diagonal entries in (4) and using the symmetry of C gives

$$\sum_j C_{ij}^2 + C_{ii} = 24 - 2u_i^2. \tag{5}$$

Lemma 4.2. *The space $\langle \mathbf{1}, u \rangle^{\perp} \subset \mathbb{R}^{12}$ has dimension 10 and is invariant under C. On it, C is diagonalizable with eigenvalues in $\{3, -4\}$. If a is the multiplicity of 3 there, then $\operatorname{tr} C = 7a - 28$.*

Proof. The vectors $\mathbf{1}$ and u are orthogonal eigenvectors of the symmetric matrix C, with eigenvalues 12 and 0. So $\langle \mathbf{1}, u \rangle^{\perp}$ is invariant. For w in this space, (4) gives $(C^2 + C)w = 12w$, so the eigenvalues there are roots of $X^2 + X - 12$. Finally $\operatorname{tr} C = 12 + 0 + 3a - 4(10 - a)$. □

5. The spectral input: C has zero diagonal

Lemma 5.1. *Let E be a finite-dimensional vector space over $\mathbb{Q}$ and T a linear map of E with $T^7 = 1$. Then $\dim E - \dim\ker(T-1)$ is divisible by 6.*

Proof. Write $X^7 - 1 = (X-1)\Phi_7(X)$ with $\Phi_7 = 1 + X + \cdots + X^6$. The two factors are coprime since $\Phi_7(1) = 7$. So $E = \ker(T-1) \oplus E'$ with $E' = \ker \Phi_7(T)$. The polynomial Φ_7 is irreducible over $\mathbb{Q}$ (Eisenstein's criterion applies to $\Phi_7(X+1)$ at the prime 7). Hence $K = \mathbb{Q}[X]/(\Phi_7)$ is a field with $[K:\mathbb{Q}] = 6$, and E' is a vector space over K, with X acting as T. Therefore $\dim_{\mathbb{Q}} E' = 6 \dim_K E'$. □

Proposition 5.2. *C has zero diagonal, and $a = 4$. Consequently*

$$\sum_j C_{ij} = 12, \qquad \sum_j C_{ij}^2 = \begin{cases} 22, & i \in \mathcal{A} \cup \mathcal{B}, \\ 24, & i \in \mathcal{M}, \end{cases} \tag{6}$$

and every entry of C is at most 4.

Proof. Let Π be the permutation matrix of s acting on $\mathbb{R}^V$, and let $F = \ker(\Pi - I)$ be the space of functions constant on $\langle s\rangle$-orbits. It has dimension 15 and is spanned by the indicator vectors $\mathbf{1}_O$ of the orbits $\{x\}$, L, R and the twelve exterior orbits. Since A commutes with Π, A preserves F, and $A|_F$ is self-adjoint for the standard inner product. Let $d = \dim(F \cap E_3)$, where $E_3 = \ker(A - 3I)$.

Step 1: $d = a + 2$. Put $\mathbf{1}_{\mathrm{ext}} = \sum_O \mathbf{1}_O$ and $e_u = \sum_O u_O \mathbf{1}_O$, where O runs over the exterior orbits. Using Lemma 2.1 and (2), (3), one checks that

$$\begin{aligned} A\mathbf{1}_x &= \mathbf{1}_L + \mathbf{1}_R, & A\mathbf{1}_L &= 7\mathbf{1}_x + \mathbf{1}_R + \mathbf{1}_{\mathrm{ext}} + e_u, \\ A\mathbf{1}_R &= 7\mathbf{1}_x + \mathbf{1}_L + \mathbf{1}_{\mathrm{ext}} - e_u, & A\mathbf{1}_{\mathrm{ext}} &= 12\mathbf{1}_L + 12\mathbf{1}_R + 12\mathbf{1}_{\mathrm{ext}}, \\ Ae_u &= 6\mathbf{1}_L - 6\mathbf{1}_R. \end{aligned}$$

For instance, Ae_u takes the value $\sum_O (1+u_O)u_O = 6$ at a vertex of L, the value $\sum_O (1-u_O)u_O = -6$ at a vertex of R, the value 0 at x, and the value $(Cu)_O = 0$ at a vertex of an exterior orbit O. So $F_0 = \langle \mathbf{1}_x, \mathbf{1}_L, \mathbf{1}_R, \mathbf{1}_{\mathrm{ext}}, e_u\rangle$ is A-invariant. On $\langle \mathbf{1}_L - \mathbf{1}_R, e_u\rangle$, A acts by $\mathbf{1}_L - \mathbf{1}_R \mapsto -(\mathbf{1}_L - \mathbf{1}_R) + 2e_u$ and $e_u \mapsto 6(\mathbf{1}_L - \mathbf{1}_R)$, with characteristic polynomial $X^2 + X - 12$, so with eigenvalues 3 and -4. On $\langle \mathbf{1}_x, \mathbf{1}_L + \mathbf{1}_R, \mathbf{1}_{\mathrm{ext}}\rangle$ the matrix of A is $\left(\begin{smallmatrix} 0 & 14 & 0 \\ 1 & 1 & 12 \\ 0 & 2 & 12 \end{smallmatrix}\right)$, with trace 13 and determinant -168. It has the eigenvalue 14 (on $\mathbf{1}$), so its other two eigenvalues have sum -1 and product -12; they are 3 and -4. Hence $A|_{F_0}$ has eigenvalues $14, 3, 3, -4, -4$. The orthogonal complement of F_0 in F consists of the vectors $\sum_O f_O \mathbf{1}_O$ supported on the exterior orbits with $f \perp \mathbf{1}$ and $f \perp u$ in $\mathbb{R}^{12}$. For such a vector, $A\sum_O f_O \mathbf{1}_O$ vanishes on x, takes the value $\sum_O (1 \pm u_O) f_O = 0$ on L and on R, and equals $\sum_{O'} C_{OO'} f_{O'}$ on O. So A acts on $F_0^{\perp} \cap F$ as C acts on $\langle \mathbf{1}, u\rangle^{\perp}$. Counting the eigenvalue 3 on both pieces gives $d = 2 + a$.

Step 2: $d \equiv 0 \pmod 6$. The matrices A and Π have rational entries. Let $E_{3,\mathbb{Q}} = \{f \in \mathbb{Q}^V : Af = 3f\}$. Its dimension over $\mathbb{Q}$ is 54, the multiplicity of the eigenvalue 3. It is invariant under Π, and $\ker(\Pi - I) \cap E_{3,\mathbb{Q}}$ has dimension d over $\mathbb{Q}$ (a kernel of a rational matrix has the same dimension over $\mathbb{Q}$ and over $\mathbb{R}$). By Lemma 5.1 applied to $T = \Pi|_{E_{3,\mathbb{Q}}}$, $54 - d \equiv 0 \pmod 6$.

Step 3. By Steps 1 and 2, $d \in \{6, 12\}$, as $2 \le d \le 12$; so $a \in \{4, 10\}$. If $a = 10$, then C acts as 3 on all of $\langle \mathbf{1}, u\rangle^{\perp}$, and the spectral decomposition of C gives

$$C = 12 \cdot \tfrac{1}{12} J + 0 \cdot \tfrac{1}{6} uu^{\mathsf{T}} + 3(I - \tfrac{1}{12} J - \tfrac{1}{6} uu^{\mathsf{T}}) = 3I + \tfrac{3}{4} J - \tfrac{1}{2} uu^{\mathsf{T}}.$$

Then $C_{mm'} = 3/4$ for distinct $m, m' \in \mathcal{M}$, which is not an integer. Hence $a = 4$, and $\operatorname{tr} C = 0$ by Lemma 4.2. Since the diagonal entries of C are nonnegative, they all vanish. Now (6) follows from (2) and (5), and $C_{ij}^2 \le 24$ gives $C_{ij} \le 4$. □

Remark 5.3. The formula $\operatorname{tr} C = 7a - 28$, equivalently $\operatorname{tr} B = 7d - 42$ for the 15×15 orbit matrix B, was used in [LM14, §8] to show $\operatorname{tr} C \in \{0, 14\}$, and in [CW25, proof of Theorem 3.11] to exclude automorphisms of order 14. The congruence $d \equiv 54 \pmod 6$ is an instance of

the character method for automorphisms of strongly regular graphs; it also appears in the computational report [Wa26], which derives the zero diagonal by a different argument.

6. Gram compression

Put

$$G = 3C + 12I - 4J - 2uu^{\mathsf{T}}. \tag{7}$$

Then $G\mathbf{1} = 36\mathbf{1} + 12\mathbf{1} - 48\mathbf{1} = 0$ and $Gu = 12u - 12u = 0$. On $\langle \mathbf{1}, u\rangle^{\perp}$, G acts as $3\cdot 3 + 12 = 21$ on the 3-eigenspace of C and as $3\cdot(-4) + 12 = 0$ on the (-4)-eigenspace. So G is 21 times the orthogonal projection onto a 4-dimensional subspace of $\mathbb{R}^{12}$, by Proposition 5.2:

$$G \succeq 0, \qquad G^2 = 21G, \qquad \operatorname{rank} G = 4. \tag{8}$$

Choose an orthonormal basis $f_1, \dots, f_4$ of the image of G and put $v_i = \sqrt{21}\,(f_1(i), \dots, f_4(i)) \in \mathbb{R}^4$ for $i = 1, \dots, 12$. Then $G_{ij} = v_i \cdot v_j$, and $v_1, \dots, v_{12}$ span $\mathbb{R}^4$. Explicitly, by Proposition 5.2,

$$|v_i|^2 = \begin{cases} 6, & i \in \mathcal{A} \cup \mathcal{B}, \\ 8, & i \in \mathcal{M}, \end{cases} \qquad v_i \cdot v_j = \begin{cases} 3C_{ij} - 6, & i,j \in \mathcal{A} \text{ or } i,j \in \mathcal{B}, \\ 3C_{ij} - 2, & i \in \mathcal{A},\ j \in \mathcal{B}, \\ 3C_{ij} - 4, & i \in \mathcal{M} \text{ or } j \in \mathcal{M}, \end{cases} \quad (i \neq j). \tag{9}$$

Since the v_i span $\mathbb{R}^4$, the identities $G\mathbf{1} = 0$ and $Gu = 0$ say that

$$\sum_i v_i = 0, \qquad \sum_{a\in\mathcal{A}} v_a = \sum_{b\in\mathcal{B}} v_b. \tag{10}$$

Write $\mathcal{A} = \{a_1, a_2, a_3\}$ and $\mathcal{B} = \{b_1, b_2, b_3\}$, and let $P = (C_{a_i a_j})$, $Q = (C_{b_i b_j})$ and $R = (C_{a_i b_j})$ be the corresponding 3×3 blocks of C. For $a \in \mathcal{A}$, row a of (3) says that $\sum_{a'\in\mathcal{A}} C_{aa'} = \sum_{b\in\mathcal{B}} C_{ab}$; we call this number σ_a. Similarly $\sigma_b = \sum_{b'\in\mathcal{B}} C_{bb'} = \sum_{a\in\mathcal{A}} C_{ab}$ for $b \in \mathcal{B}$. In matrix form,

$$P\mathbf{1} = R\mathbf{1}, \qquad Q\mathbf{1} = R^{\mathsf{T}}\mathbf{1}. \tag{11}$$

In particular the sum of the three off-diagonal entries $C_{aa'}$ ($\{a, a'\} \subset \mathcal{A}$) equals the sum of the three entries $C_{bb'}$ ($\{b, b'\} \subset \mathcal{B}$), since both are half the sum of the entries of R. We call this common value h. We think of P and Q as edge-weighted triangles on $\mathcal{A}$ and on $\mathcal{B}$.

Relabelling $\mathcal{A}$ and $\mathcal{B}$ separately (that is, permuting the first three and the next three indices) preserves u, and therefore preserves (2)–(4) and Proposition 5.2. So does exchanging $\mathcal{A}$ with $\mathcal{B}$, which replaces u by $-u$. We use these relabellings freely.

7. The LL/RR block

Lemma 7.1. *$C_{aa'} \geq 1$ for distinct $a, a' \in \mathcal{A}$, and $C_{bb'} \geq 1$ for distinct $b, b' \in \mathcal{B}$.*

Proof. If $C_{aa'} = 0$, then $|v_a + v_{a'}|^2 = 6 + 6 - 12 = 0$, so $v_{a'} = -v_a$. For $b \in \mathcal{B}$ this gives $(3C_{ab} - 2) + (3C_{a'b} - 2) = 0$, that is, $3(C_{ab} + C_{a'b}) = 4$, which is impossible. The same argument works in $\mathcal{B}$. □

Lemma 7.2. *$h \in \{3, 4, 5\}$.*

Proof. Let $e_{\mathcal{A}}$ and $e_{\mathcal{B}}$ be the indicator vectors of $\mathcal{A}$ and $\mathcal{B}$ in $\mathbb{R}^{12}$, and put $t = Ce_{\mathcal{A}}$. By (3), $t = Ce_{\mathcal{B}}$ as well. Then $\sum_{i\in\mathcal{A}} t_i = e_{\mathcal{A}}^{\mathsf{T}} C e_{\mathcal{A}} = 2h$ and $\sum_{i\in\mathcal{B}} t_i = e_{\mathcal{B}}^{\mathsf{T}} C e_{\mathcal{B}} = 2h$, while $\sum_{i\in\mathcal{M}} t_i = \mathbf{1}^{\mathsf{T}} C e_{\mathcal{A}} - 4h = 36 - 4h$ by (2). By (4),

$$|t|^2 = e_{\mathcal{A}}^{\mathsf{T}} C^2 e_{\mathcal{A}} = e_{\mathcal{A}}^{\mathsf{T}}(12I + 12J - 2uu^{\mathsf{T}} - C)e_{\mathcal{A}} = 36 + 108 - 18 - 2h = 126 - 2h.$$

The Cauchy–Schwarz inequality on the three blocks $\mathcal{A}, \mathcal{B}, \mathcal{M}$, of sizes $3, 3, 6$, gives

$$126 - 2h \;\geq\; \tfrac{(2h)^2}{3} + \tfrac{(2h)^2}{3} + \tfrac{(36-4h)^2}{6}.$$

Multiplying by 6 and simplifying gives $8h^2 - 69h + 135 \leq 0$, that is, $(h-3)(8h-45) \leq 0$. So $3 \leq h \leq 45/8$. □

By Lemmas 7.1 and 7.2, the multiset of edge weights of P (and of Q) is one of $\{1,1,1\}$ ($h=3$), $\{1,1,2\}$ ($h=4$), $\{1,1,3\}$ or $\{1,2,2\}$ ($h=5$). After relabelling $\mathcal{A}$ and $\mathcal{B}$ separately, each of P and Q is one of

$$T_{111}=\begin{pmatrix}0&1&1\\1&0&1\\1&1&0\end{pmatrix},\ T_{112}=\begin{pmatrix}0&1&1\\1&0&2\\1&2&0\end{pmatrix},\ T_{113}=\begin{pmatrix}0&1&1\\1&0&3\\1&3&0\end{pmatrix},\ T_{122}=\begin{pmatrix}0&1&2\\1&0&2\\2&2&0\end{pmatrix}.$$

Their row sums are $(2,2,2)$, $(2,3,3)$, $(2,4,4)$ and $(3,3,4)$.

7.1. **The case $h=3$.**

Proposition 7.3. *If $h=3$, then $P=Q=T_{111}$, $\sum_{\mathcal{A}} v_a=\sum_{\mathcal{B}} v_b=0$, and every row and column of R sums to 2. After relabelling, R is one of*

$$R_1=2I,\qquad R_2=\begin{pmatrix}2&0&0\\0&1&1\\0&1&1\end{pmatrix},\qquad R_3=T_{111}.$$

Moreover every $m\in\mathcal{M}$ satisfies $\sum_{a\in\mathcal{A}} C_{ma}=\sum_{b\in\mathcal{B}} C_{mb}=4$.

Proof. By (9), $|\sum_{\mathcal{A}} v_a|^2=3\cdot 6+6\cdot(3-6)=0$, so $\sum_{\mathcal{A}} v_a=0$, and $\sum_{\mathcal{B}} v_b=0$ by (10). Taking the inner product of $\sum_{\mathcal{A}} v_a=0$ with v_b gives $\sum_a(3R_{ab}-2)=0$, so the column sums of R are 2; the row sums are treated in the same way. The inner product with v_m gives $\sum_a(3C_{ma}-4)=0$. A nonnegative integral 3×3 matrix with all line sums 2 is a sum $\pi_1+\pi_2$ of two permutation matrices (König's theorem). Permuting the rows, we may take $\pi_1=I$; conjugating by a permutation matrix (that is, permuting rows and columns simultaneously), π_2 becomes the identity, a transposition or a 3-cycle. The first two give R_1 and R_2. The third gives $I+Z$ with Z the matrix of a 3-cycle. Since $I+Z+Z^2=J$, we have $I+Z=J-Z^2$, and permuting the columns by Z^{-2} turns this into $J-I=R_3$. All these row and column permutations are relabellings of $\mathcal{A}$ and $\mathcal{B}$, and they preserve $P=Q=T_{111}$. □

7.2. **The cases $h=4,5$: residuals.** Now $P\in\{T_{112},T_{113},T_{122}\}$. The Gram matrix $G_{\mathcal{A}}=(v_a\cdot v_{a'})_{a,a'\in\mathcal{A}}$ has diagonal entries 6 and off-diagonal entries $3C_{aa'}-6\in\{-3,0,3\}$. Its determinant is 108, 108 and 162 for $P=T_{112},T_{113},T_{122}$. So v_{a_1},v_{a_2},v_{a_3} span a 3-dimensional subspace $U\subset\mathbb{R}^4$. Fix a unit vector $\nu\perp U$, and for $b\in\mathcal{B}$ put $z_b=v_b\cdot\nu$. Let $c=(R_{a_1b},R_{a_2b},R_{a_3b})$ be the column of R at b; its entry sum is σ_b by (11). The inner products of v_b with v_{a_1},v_{a_2},v_{a_3} form the vector $X=3c-\mathbf{21}$, so the orthogonal projection of v_b to U has squared length $X^{\mathsf{T}}G_{\mathcal{A}}^{-1}X$. Hence

$$z_b^2=6-X^{\mathsf{T}}G_{\mathcal{A}}^{-1}X,\qquad z_{b_1}+z_{b_2}+z_{b_3}=0, \tag{12}$$

the second identity because $\sum_{\mathcal{B}} v_b=\sum_{\mathcal{A}} v_a\in U$ by (10). Put $\alpha_b=12z_b^2$.

Lemma 7.4. *Let $c=(c_1,c_2,c_3)$ be the column of R at some $b\in\mathcal{B}$, and $s=c_1+c_2+c_3$. Then*

$$P=T_{112}:\quad \alpha_b=72s-18s^2-8+6c_1(4-3c_1)-9(c_2-c_3)^2,$$
$$P=T_{113}:\quad \alpha_b=28+36s-9s^2+6c_1(4-3c_1)-18(c_2-c_3)^2,$$
$$P=T_{122}:\quad \alpha_b=40+48p-18p^2-6q^2-2(3c_3-2)^2,\qquad p=c_1+c_2,\ q=c_1-c_2.$$

In particular α_b is an integer.

Proof. The inverses are

$$12\,G_{\mathcal{A}}^{-1}=\begin{pmatrix}4&2&2\\2&3&1\\2&1&3\end{pmatrix},\quad \begin{pmatrix}3&1&1\\1&3&-1\\1&-1&3\end{pmatrix},\quad \frac{2}{3}\begin{pmatrix}4&2&0\\2&4&0\\0&0&3\end{pmatrix}$$

for $P=T_{112},T_{113},T_{122}$; one checks directly that the product with $G_{\mathcal{A}}$ is $12I$. Write $M=12G_{\mathcal{A}}^{-1}$, so that $\alpha_b=72-X^{\mathsf{T}}MX$ with $X=3c-\mathbf{21}$. For T_{112}, $M\mathbf{1}=(8,6,6)^{\mathsf{T}}$, $\mathbf{1}^{\mathsf{T}}M\mathbf{1}=20$ and $c^{\mathsf{T}}Mc=2s^2+2c_1^2+(c_2-c_3)^2$. Hence $X^{\mathsf{T}}MX=9c^{\mathsf{T}}Mc-12(8c_1+6c_2+6c_3)+80$, which

gives the formula. For T_{113}, $M\mathbf{1} = (5,3,3)^\mathsf{T}$, $\mathbf{1}^\mathsf{T}M\mathbf{1} = 11$ and $c^\mathsf{T}Mc = s^2 + 2c_1^2 + 2(c_2 - c_3)^2$, and the computation is the same. For T_{122}, $X^\mathsf{T}MX = \frac{8}{3}(X_1^2 + X_1X_2 + X_2^2) + 2X_3^2$, and $X_1^2 + X_1X_2 + X_2^2 = 9(c_1^2 + c_1c_2 + c_2^2) - 18(c_1 + c_2) + 12 = \frac{9}{4}(3p^2 + q^2) - 18p + 12$. □

Table 1 in Appendix A lists α for every column c with the column sums needed below, computed from Lemma 7.4. The columns with $\alpha \geq 0$ are the following.

P	s	columns c with $\alpha \geq 0$, and their value of α
T_{112}	2	$(0,1,1)$: 64; $(1,0,1),(1,1,0)$: 61; $(2,0,0)$: 40; $(0,0,2),(0,2,0)$: 28
	3	$(1,1,1)$: 52; $(0,1,2),(0,2,1)$: 37; $(1,0,2),(1,2,0)$: 16; $(2,0,1),(2,1,0)$: 13
T_{113}	2	$(0,1,1)$: 64; $(1,0,1),(1,1,0)$: 52; $(2,0,0)$: 40
	3	$(1,1,1)$: 61; $(0,1,2),(0,2,1)$: 37; $(2,0,1),(2,1,0)$: 13
	4	$(0,2,2)$: 28; $(1,1,2),(1,2,1)$: 16; $(2,1,1)$: 4
T_{122}	3	$(1,1,1)$: 62; $(0,2,1),(2,0,1)$: 38; $(0,1,2),(1,0,2)$: 32; $(1,2,0),(2,1,0)$: 8
	4	$(1,1,2)$: 32; $(1,2,1),(2,1,1)$: 14; $(0,2,2),(2,0,2)$: 8

Since $z_b^2 \geq 0$, every column of R appears in this list. None of the listed values is 0.

Three real numbers with zero sum have the property that the one with the largest absolute value is, in absolute value, the sum of the other two. So by (12), after renaming the three elements of $\mathcal{B}$ as j,k,l suitably,

$$\sqrt{\alpha_l} = \sqrt{\alpha_j} + \sqrt{\alpha_k}, \qquad \text{equivalently} \qquad \alpha_l = \alpha_j + \alpha_k + 2\sqrt{\alpha_j\alpha_k}, \tag{13}$$

and in particular $\alpha_j\alpha_k$ is a perfect square.

Proposition 7.5. *If $h \in \{4,5\}$, then up to relabelling $Q = P$, and (P,R) is one of*

$$(T_{112}, T_{112}), \qquad \left(T_{113}, \begin{pmatrix} 0 & 1 & 1 \\ 1 & 1 & 2 \\ 1 & 2 & 1 \end{pmatrix}\right), \qquad \left(T_{122}, \begin{pmatrix} 0 & 1 & 2 \\ 1 & 2 & 0 \\ 2 & 0 & 2 \end{pmatrix}\right).$$

Proof. The column sums of R are the row sums of Q, by (11), and the row sums of R are those of P. For each pair of elements of $\mathcal{B}$ whose values of α have a square product, (13) gives the value $(\sqrt{\alpha_j} + \sqrt{\alpha_k})^2$ required at the third element.

$P = Q = T_{112}$. The column at b_1 has sum 2 and those at b_2, b_3 have sum 3. So $\alpha_{b_1} \in \{64, 61, 40, 28\}$ and $\alpha_{b_2}, \alpha_{b_3} \in \{52, 37, 16, 13\}$. The square products of an element of $\{52, 37, 16, 13\}$ with an element of the same set are $16 \cdot 16$, $13 \cdot 13$, $13 \cdot 52$, $52 \cdot 52$ and $37 \cdot 37$; they require $\alpha_{b_1} = 64, 52, 117, 208, 148$ respectively. The only square product of an element of $\{64, 61, 40, 28\}$ with an element of $\{52, 37, 16, 13\}$ is $64 \cdot 16$, which requires the value 144 at the remaining element of $\{b_2, b_3\}$. So the only possibility is $\sqrt{64} = \sqrt{16} + \sqrt{16}$, that is, $c_{b_1} = (0,1,1)$ and $c_{b_2}, c_{b_3} \in \{(1,0,2),(1,2,0)\}$. The second row of R sums to 3, so the two latter columns are different. The transposition $b_2 \leftrightarrow b_3$ preserves $Q = T_{112}$, so we may take $R = T_{112}$.

$P = Q = T_{113}$. Now $\alpha_{b_1} \in \{64, 52, 40\}$ (sum 2) and $\alpha_{b_2}, \alpha_{b_3} \in \{28, 16, 4\}$ (sum 4). The square products within $\{28, 16, 4\}$ are $16 \cdot 16$, $4 \cdot 4$, $16 \cdot 4$ and $28 \cdot 28$; they require $\alpha_{b_1} = 64, 16, 36, 112$. The square products between $\{64, 52, 40\}$ and $\{28, 16, 4\}$ are $64 \cdot 16$ and $64 \cdot 4$; they require 144 and 100 in $\{28, 16, 4\}$. So the only possibility is again $\sqrt{64} = \sqrt{16} + \sqrt{16}$: $c_{b_1} = (0,1,1)$ and $c_{b_2}, c_{b_3} \in \{(1,1,2),(1,2,1)\}$. The second row of R sums to 4, so these two columns are different, and swapping b_2, b_3 gives the stated R.

$P = T_{113}$, $Q = T_{122}$. The row sums of Q are $(3,3,4)$, so $\alpha_{b_1}, \alpha_{b_2} \in \{61, 37, 13\}$ and $\alpha_{b_3} \in \{28, 16, 4\}$. The numbers $61, 37, 13$ are distinct primes, and none of them divides an element of $\{28, 16, 4\}$. So the only square products are $61^2, 37^2, 13^2$, for the pair $\{b_1, b_2\}$; they require $\alpha_{b_3} = 244, 148, 52$. No R exists. Exchanging $\mathcal{A}$ and $\mathcal{B}$ disposes of $P = T_{122}$, $Q = T_{113}$.

$P = Q = T_{122}$. The row sums of Q are $(3,3,4)$, so $\alpha_{b_1}, \alpha_{b_2} \in \{62, 38, 32, 8\}$ and $\alpha_{b_3} \in \{32, 14, 8\}$. The square products for the pair $\{b_1, b_2\}$ are $62 \cdot 62$, $38 \cdot 38$, $32 \cdot 32$, $8 \cdot 8$ and $8 \cdot 32$; they require $\alpha_{b_3} = 248, 152, 128, 32, 72$. For a pair $\{b_j, b_3\}$ with $j \leq 2$, the square products occur for $(\alpha_{b_j}, \alpha_{b_3}) = (32,32)$, $(8,8)$, $(8,32)$ and $(32,8)$; they require 128, 32, 72, 72

at the remaining element of $\{b_1, b_2\}$, and only 32 is available there. So the only identity is $\sqrt{32} = \sqrt{8} + \sqrt{8}$. If the value 32 sits at b_3, then $c_{b_3} = (1,1,2)$ and $c_{b_1}, c_{b_2} \in \{(1,2,0),(2,1,0)\}$, and the third row of R sums to $2 \neq 4$. So the value 32 sits at b_1 or b_2, and by the symmetry $b_1 \leftrightarrow b_2$ of Q we may take it at b_1. Then $c_{b_1} \in \{(0,1,2),(1,0,2)\}$, $c_{b_2} \in \{(1,2,0),(2,1,0)\}$ and $c_{b_3} \in \{(0,2,2),(2,0,2)\}$. Write these as $(\epsilon, 1-\epsilon, 2)$, $(1+\eta, 2-\eta, 0)$ and $(2\theta, 2-2\theta, 2)$ with $\epsilon, \eta, \theta \in \{0,1\}$. The first row of R sums to 3, so $\epsilon + \eta + 2\theta = 2$. Either $\theta = 1$ and $\epsilon = \eta = 0$, which is the stated R, or $\theta = 0$ and $\epsilon = \eta = 1$, which gives the stated R after the transposition $a_1 \leftrightarrow a_2$, a symmetry of P. □

Theorem 7.6 (the six configurations)**.** *After relabelling $\mathcal{A}$ and $\mathcal{B}$, and possibly exchanging them, the 6×6 block of C on $\mathcal{A} \cup \mathcal{B}$ is one of the following. In every case $Q = P$ and $R = R^{\mathsf{T}}$.*

case	h	$P = Q$	R
1	*3*	T_{111}	$2I$
2	*3*	T_{111}	$\left(\begin{smallmatrix} 2&0&0\\ 0&1&1\\ 0&1&1 \end{smallmatrix}\right)$
3	*3*	T_{111}	T_{111}
4	*4*	T_{112}	T_{112}
5	*5*	T_{113}	$\left(\begin{smallmatrix} 0&1&1\\ 1&1&2\\ 1&2&1 \end{smallmatrix}\right)$
6	*5*	T_{122}	$\left(\begin{smallmatrix} 0&1&2\\ 1&2&0\\ 2&0&2 \end{smallmatrix}\right)$

Proof. Combine Propositions 7.3 and 7.5. □

8. The mixed orbits

Fix one of the six cases of Theorem 7.6. For $i = 1,2,3$ put

$$w_i^- = v_{a_i} - v_{b_i}, \qquad w_i^+ = v_{a_i} + v_{b_i},$$

and let $\mathsf{W}^- = \langle w_1^-, w_2^-, w_3^- \rangle$ and $\mathsf{W}^+ = \langle w_1^+, w_2^+, w_3^+ \rangle$.

Lemma 8.1. $\mathbb{R}^4 = \mathsf{W}^- \oplus \mathsf{W}^+$*, an orthogonal direct sum. The Gram matrices of (w_i^-) and (w_i^+) are*

$$G^- = 2(3P - 3R + 12I - 4J), \qquad G^+ = 2(3P + 3R + 12I - 8J),$$

listed for the six cases in Table 2 (Appendix A).

Proof. We have $w_i^- \cdot w_j^+ = (v_{a_i} \cdot v_{a_j} - v_{b_i} \cdot v_{b_j}) + (v_{a_i} \cdot v_{b_j} - v_{b_i} \cdot v_{a_j})$. By (9), the first bracket vanishes because $Q = P$, and the second because $R = R^{\mathsf{T}}$. The Gram matrices follow from (9). From Table 2, $\operatorname{rank} G^- + \operatorname{rank} G^+ = 4$ in each case. Since $\mathsf{W}^- \perp \mathsf{W}^+$, the sum $\mathsf{W}^- + \mathsf{W}^+$ is direct and has dimension $\operatorname{rank} G^- + \operatorname{rank} G^+ = 4$, so it is $\mathbb{R}^4$. □

Now let $m \in \mathcal{M}$. Write $x_i = C_{ma_i}$ and $y_i = C_{mb_i}$ (integers in $[0,4]$), and put $d_i = x_i - y_i$, $n_i = x_i + y_i$ and $e_i = 3n_i - 8$. By (9),

$$v_m \cdot w_i^- = 3d_i, \qquad v_m \cdot w_i^+ = e_i, \qquad 8 = |v_m|^2 = |\pi^- v_m|^2 + |\pi^+ v_m|^2, \tag{14}$$

where $\pi^\pm$ is the orthogonal projection onto $\mathsf{W}^\pm$. Row m of (3) gives $d_1 + d_2 + d_3 = 0$. Note that $d_i \equiv n_i \pmod 2$, $|d_i| \le n_i$ and $e_i \equiv 1 \pmod 3$. We use two facts from Euclidean geometry. If $g_1, \dots, g_r$ is an orthogonal basis of a subspace, the projection π onto it satisfies $|\pi z|^2 = \sum_j (z \cdot g_j)^2 / |g_j|^2$. If f_1, f_2, f_3 span a plane and have Gram matrix $c(3I - J)$ for some $c > 0$ (a *triangle frame*), then $f_1 + f_2 + f_3 = 0$, and $\sum_i (z \cdot f_i) f_i = 3c\, z$ for z in the plane (check it on $z = f_1$ and $z = f_2$), so $|\pi z|^2 = \frac{1}{3c} \sum_i (z \cdot f_i)^2$ for the projection π onto the plane.

Case 1. Here $G^- = 2(3I - J)$ and $G^+ = 10(3I - J)$ are triangle frames. So (14) gives $\frac{3}{2}\sum d_i^2 + \frac{1}{30}\sum e_i^2 = 8$. Since $h = 3$, Proposition 7.3 gives $\sum n_i = 8$, so $\sum e_i = 0$. Put $k_i = n_i - 3 \geq -3$, so that $e_i = 3k_i + 1$, $\sum k_i = -1$ and $\sum e_i^2 = 9\sum k_i^2 - 3$. Then

$$5\sum d_i^2 + \sum k_i^2 = 27, \qquad k_i \equiv d_i + 1 \pmod 2.$$

Since $\sum d_i = 0$, either all d_i are even or exactly two are odd.

If all d_i are even, then $\sum d_i^2 \leq 5$ forces $d = 0$, and all k_i are odd with $\sum k_i^2 = 27$. The sums of three odd squares equal to 27 are $25 + 1 + 1$ and $9 + 9 + 9$, and no choice of signs gives $\sum k_i = -1$.

If exactly two d_i are odd, then $\sum d_i^2 \leq 5$ forces d to be a permutation of $(0, 1, -1)$, so $\sum k_i^2 = 17$, with k_i odd exactly where $d_i = 0$. The decompositions of 17 as an odd square plus two even squares are $1 + 16 + 0$ and $9 + 4 + 4$. With $\sum k_i = -1$, the first gives no solution ($\pm 1 \pm 4 + 0$ is odd but never -1) and the second gives only $k = (3, -2, -2)$, with 3 where $d_i = 0$. Hence n is a permutation of $(6, 1, 1)$, with 6 where $d_i = 0$, and

$$(x; y) \in \{(3, 1, 0; 3, 0, 1),\ (3, 0, 1; 3, 1, 0)\} \quad \text{up to a common permutation of the indices.}$$

In particular $C_{ma_1} \in \{0, 1, 3\}$.

Case 2. The vectors w_1^-, $w_2^- - w_3^-$ form an orthogonal basis of W^-, with squared norms 4 and 36, and w_1^+, $w_2^+ - w_3^+$ form an orthogonal basis of W^+, with squared norms 20 and 36. So $\frac{9}{4}d_1^2 + \frac{1}{4}(d_2 - d_3)^2 + \frac{1}{20}e_1^2 + \frac{1}{4}(n_2 - n_3)^2 = 8$, that is,

$$45d_1^2 + 5(d_2 - d_3)^2 + e_1^2 + 5(n_2 - n_3)^2 = 160.$$

Hence $5 \mid e_1 = 3n_1 - 8$, so $n_1 \in \{1, 6\}$. Also $\sum n_i = 8$ by Proposition 7.3. If $n_1 = 6$, then d_1 is even and $45d_1^2 \leq 60$, so $d_1 = 0$; then $(d_2 - d_3)^2 + (n_2 - n_3)^2 = 12$, but $n_2 + n_3 = 2$ makes the left side at most $4 + 4 = 8$. If $n_1 = 1$, then $d_1 = \pm 1$ and $(d_2 - d_3)^2 + (n_2 - n_3)^2 = 18$ with $n_2 + n_3 = 7$. Both $n_2 - n_3$ and $d_2 - d_3 \equiv d_2 + d_3 = -d_1$ are odd, so both squares equal 9, and $\{n_2, n_3\} = \{2, 5\}$. Solving with $d_2 + d_3 = -d_1$ and $d_i \equiv n_i \pmod 2$ gives

$$(x; y) \in \{(1, 3, 0; 0, 2, 2),\ (1, 0, 3; 0, 2, 2),\ (0, 2, 2; 1, 3, 0),\ (0, 2, 2; 1, 0, 3)\}.$$

So $C_{ma_1} \in \{0, 1\}$.

Case 3. Here $G^- = 8(3I - J)$ and $G^+ = 4(3I - J)$ are triangle frames, so $\frac{3}{8}\sum d_i^2 + \frac{1}{12}\sum e_i^2 = 8$. With $e_i = 3k_i + 1$ and $\sum k_i = -1$ as in Case 1,

$$\sum d_i^2 + 2\sum k_i^2 = 22.$$

If exactly two d_i are odd, then $\sum d_i^2 \equiv 2 \pmod 4$; exactly one k_i is odd, so $2\sum k_i^2 \equiv 2 \pmod 4$, and the left side is divisible by 4, a contradiction. So all d_i are even and all k_i are odd. A sum of three odd squares is $\equiv 3 \pmod 8$, so $\sum k_i^2 \in \{3, 11\}$, and then $\sum d_i^2 \in \{16, 0\}$. All d_i even with zero sum gives $\sum d_i^2 \equiv 0 \pmod 8$, and $\sum d_i^2 = 16$ would need $d_i = 2f_i$ with $f_1^2 + f_1 f_2 + f_2^2 = 2$, which is impossible (this form is odd unless f_1, f_2 are both even). So $d = 0$ and $\{k_i^2\} = \{9, 1, 1\}$; with $\sum k_i = -1$ this means k is a permutation of $(-3, 1, 1)$. Thus $x = y$ is a permutation of $(0, 2, 2)$, and $C_{ma_1} \in \{0, 2\}$.

Case 4. Here $G^- = 8(3I - J)$. In G^+ we have $|w_2^+ - w_3^+|^2 = 0$, so $w_2^+ = w_3^+$ and therefore $e_2 = e_3$, that is, $n_2 = n_3$. The vectors w_1^+ and $w_1^+ + 2w_2^+$ form an orthogonal basis of W^+, with squared norms 8 and 24. Moreover $v_m \cdot (w_1^+ + 2w_2^+) = e_1 + e_2 + e_3 = 6\sigma - 24$, where $\sigma = \sum x_i = \sum y_i$. Hence

$$3\sum d_i^2 + e_1^2 + 12(\sigma - 4)^2 = 64.$$

From $n_2 = n_3$ we get $d_2 \equiv d_3 \pmod 2$, so $d_1 = -(d_2 + d_3)$ and n_1 are even. As $e_1^2 \leq 64$, $n_1 \in \{0, 2, 4\}$. If all d_i are even, write $d_i = 2f_i$; then $\sum d_i^2 = 8(f_1^2 + f_1 f_2 + f_2^2)$, and $f_1^2 + f_1 f_2 + f_2^2$ is odd or divisible by 4. So $\sum d_i^2 \in \{0, 8, 24, 32, \dots\}$ in that case, and $\sum d_i^2 \equiv 2 \pmod 4$ otherwise.

- $n_1 = 0$: then $e_1^2 = 64$, so $d = 0$ and $\sigma = 4$; with $n_2 = n_3$ this gives $x = y = (0, 2, 2)$.

- $n_1 = 2$: then $\sum d_i^2 + 4(\sigma-4)^2 = 20$. So $\sum d_i^2 \equiv 0 \pmod 4$, all d_i are even, and $\sum d_i^2 \in \{0, 8\}$. Then $(\sigma-4)^2 \in \{5, 3\}$, which is impossible.
- $n_1 = 4$: then $\sum d_i^2 + 4(\sigma-4)^2 = 16$. As before $\sum d_i^2 \in \{0, 8\}$, and $(\sigma-4)^2 = 2$ is impossible, so $d = 0$ and $\sigma \in \{2, 6\}$. With $x_1 = 2$ and $x_2 = x_3$ this gives $x = y \in \{(2,0,0), (2,2,2)\}$.

So $C_{ma_1} \in \{0, 2\}$.

Case 5. The vectors w_1^-, $w_2^- - w_3^-$ form an orthogonal basis of W^-, with squared norms 16 and 24. In W^+ we have $w_2^+ = w_3^+$, so $n_2 = n_3$, and w_1^+, $w_1^+ + 2w_2^+$ form an orthogonal basis, with squared norms 8 and 48. With σ as in Case 4,

$$9d_1^2 + 6(d_2 - d_3)^2 + 2e_1^2 + 12(\sigma-4)^2 = 128.$$

As in Case 4, d_1 and n_1 are even, and $2e_1^2 \le 128$ gives $n_1 \in \{0, 2, 4\}$. Also $d_2 - d_3$ is even, so $(d_2-d_3)^2 \in \{0, 4, 16\}$, the bound $6(d_2-d_3)^2 \le 128$ excluding larger values.

- $n_1 = 0$: then $2e_1^2 = 128$, so $d_1 = 0$, $d_2 = d_3$, hence $d = 0$, and $\sigma = 4$. So $x = y = (0, 2, 2)$.
- $n_1 = 2$: then $3d_1^2 + 2(d_2-d_3)^2 + 4(\sigma-4)^2 = 40$ with $d_1 \in \{0, \pm 2\}$. For $d_1 = 0$ this needs $(\sigma-4)^2 \in \{10, 8, 2\}$, and for $d_1 = \pm 2$ it needs $(\sigma-4)^2 \in \{7, 5, -1\}$. None is a square.
- $n_1 = 4$: then $3d_1^2 + 2(d_2-d_3)^2 + 4(\sigma-4)^2 = 32$ with $d_1 \in \{0, \pm 2\}$ (as $3d_1^2 \le 32$). For $d_1 = \pm 2$ this needs $(\sigma-4)^2 \in \{5, 3, -3\}$, impossible. For $d_1 = 0$ it needs $(\sigma-4)^2 \in \{8, 6, 0\}$, so $(d_2-d_3)^2 = 16$ and $\sigma = 4$. With $d_2 + d_3 = 0$, $x_1 = y_1 = 2$ and $x_2 + x_3 = y_2 + y_3 = 2$, this gives $(x; y) \in \{(2,2,0;2,0,2),\ (2,0,2;2,2,0)\}$.

So $C_{ma_1} \in \{0, 2\}$.

Case 6. Here G^- has rank 1: $w_2^- = w_3^- = -\frac{1}{2}w_1^-$ and $|w_1^-|^2 = 16$. So $d_2 = d_3 = -d_1/2$ and $|\pi^- v_m|^2 = \frac{9}{16}d_1^2$. The matrix G^+ has rank 3, and w_1^+, $w_1^+ + 2w_2^+$, $w_3^+ - w_1^+$ form an orthogonal basis of W^+, with squared norms 8, 72 and 12. Since $v_m \cdot (w_1^+ + 2w_2^+) = 3(n_1 + 2n_2 - 8)$ and $v_m \cdot (w_3^+ - w_1^+) = 3(n_3 - n_1)$,

$$9d_1^2 + 2(3n_1 - 8)^2 + 2(n_1 + 2n_2 - 8)^2 + 12(n_3 - n_1)^2 = 128.$$

Here $d_1 = -2d_2$ is even, and $9d_1^2 \le 128$ gives $d_1 \in \{0, \pm 2\}$.

- $d_1 = 0$: then $d = 0$, so $x = y$ and $n = 2x$, and the equation becomes $(3x_1 - 4)^2 + (x_1 + 2x_2 - 4)^2 + 6(x_3 - x_1)^2 = 16$. So $x_1 \le 2$. If $x_1 = 0$, then $x = (0, 2, 0)$. If $x_1 = 1$, then $(2x_2 - 3)^2 + 6(x_3 - 1)^2 = 15$; the odd square $(2x_2 - 3)^2$ is 1 or 9, and only 9 works, so $x_2 \in \{0, 3\}$ and $x_3 \in \{0, 2\}$. If $x_1 = 2$, then $2(x_2 - 1)^2 + 3(x_3 - 2)^2 = 6$, which has no solution.
- $d_1 = \pm 2$: then n_1 is even, $d_2 = d_3 = \mp 1$, so n_2, n_3 are odd, and $(3n_1 - 8)^2 + (n_1 + 2n_2 - 8)^2 + 6(n_3 - n_1)^2 = 46$ with $n_3 - n_1$ odd. So $6(n_3 - n_1)^2 = 6$, and $|3n_1 - 8| \le 6$ with $n_1 \ge |d_1| = 2$ gives $n_1 \in \{2, 4\}$. For $n_1 = 2$ the equation forces $(2n_2 - 6)^2 = 36$, so $n_2 \in \{0, 6\}$ is even, a contradiction. For $n_1 = 4$ it forces $(2n_2 - 4)^2 = 24$, which is impossible.

So $x = y$ and $C_{ma_1} = x_1 \in \{0, 1\}$. The equation has the five solutions

$$x = y \in \{(0,2,0),\ (1,0,0),\ (1,0,2),\ (1,3,0),\ (1,3,2)\};$$

the last one violates (6), but this is not needed.

Conclusion. Now consider row a_1 of C. Its entries in the columns $\mathcal{A} \cup \mathcal{B}$ are given by Theorem 7.6. By (6), its six entries in the columns $\mathcal{M}$ sum to $12 - 2\sigma_{a_1}$, and their squares sum to $22 - \sum_{j \in \mathcal{A}\cup\mathcal{B}} C_{a_1 j}^2$. By the six cases above, each of these entries $C_{a_1 m} = C_{m a_1}$ lies in a set of two or three values:

case	row a_1 on $\mathcal{A}\,\vert\,\mathcal{B}$	values of C_{a_1m}	$\sum_m C_{a_1m}$	$\sum_m C_{a_1m}^2$	why impossible
1	$0,1,1\,\vert\,2,0,0$	$\{0,1,3\}$	8	16	$N_1+3N_3=8$, $N_1+9N_3=16$
2	$0,1,1\,\vert\,2,0,0$	$\{0,1\}$	8	16	the sum is at most 6
3	$0,1,1\,\vert\,0,1,1$	$\{0,2\}$	8	18	$c^2=2c$ gives square-sum 16
4	$0,1,1\,\vert\,0,1,1$	$\{0,2\}$	8	18	as in case 3
5	$0,1,1\,\vert\,0,1,1$	$\{0,2\}$	8	18	as in case 3
6	$0,1,2\,\vert\,0,1,2$	$\{0,1\}$	6	12	$c^2=c$ gives square-sum 6

In case 1, N_1 and N_3 denote the numbers of entries equal to 1 and to 3; the two equations give $6N_3=8$, which is impossible. In case 2, six entries in $\{0,1\}$ cannot sum to 8. Every case is contradictory. So no matrix C as in Propositions 4.1 and 5.2 exists, and this proves Theorem 1.1. □

9. Remarks

What the proof uses. Besides the parameters $(99,14,1,2)$, the proof uses one automorphism s of order 7. From the graph it uses only Lemma 2.1, the orbit description of Section 3 and the rationality of the eigenspaces. From representation theory it uses only the irreducibility of the cyclotomic polynomial Φ_7 over $\mathbb{Q}$. Everything after Proposition 5.2 takes place in the 12×12 matrix C. The only facts about C used there are those of Propositions 4.1 and 5.2. Nothing in Sections 6 to 8 uses a computer. In Section 7, every value of α follows from the closed formulas of Lemma 7.4 and is listed in Table 1.

Independent machine checks. The following exact-arithmetic computations were run as sanity checks. They are not part of the proof. The code is in the directory `checks/` of the repository [Dj26].

- All symmetric 6×6 blocks with zero diagonal and entries in $\{0,\dots,4\}$ that satisfy (11), the bounds coming from (6), and the condition that the corresponding 6×6 block of G is positive semidefinite of rank at most 4. They fall into 8 classes up to relabelling and exchange of $\mathcal{A}$ and $\mathcal{B}$. Six classes are those of Theorem 7.6. The other two have $h=6$ and are excluded by Lemma 7.2. Lemma 7.1 is not assumed in this computation.
- For each of the six configurations, all $(x;y)\in\{0,\dots,4\}^6$ such that the 7×7 block of G on $\mathcal{A}\cup\mathcal{B}\cup\{m\}$ is positive semidefinite of rank at most 4 and that satisfy $\sum x_i=\sum y_i$ and the bounds from (6). They are exactly the vectors found in Section 8, except that in Case 6 the vector $(1,3,2;1,3,2)$ is excluded by (6). The final contradiction was also confirmed.
- The formulas of Lemma 7.4 were compared with direct evaluation of $12(6-X^{\mathsf{T}}G_{\mathcal{A}}^{-1}X)$ for all $c\in\{0,\dots,4\}^3$.

Formal verification. Theorem 1.1 has been formalized in Lean 4 [dMU21] with Mathlib [Mat20], version `v4.35.0-rc2`, with no `sorry`; the sources are in the directory `lean/` of the repository [Dj26]. The statement is isolated in a file that imports only Mathlib and contains no definition. For G with `G.IsSRGWith 99 14 1 2` (Mathlib's definition of a strongly regular graph) and an automorphism σ with $\sigma^7=1$, it asserts $\sigma=1$. It is also stated in group form ($\operatorname{Aut}\Gamma$ has no element of order 7, and $7\nmid|\operatorname{Aut}\Gamma|$), and in an elementary form that uses no graph library: a Boolean adjacency matrix on $\{0,\dots,98\}$, with the counts given by `List.countP`. As a test of non-vacuity, the same formulas with $(21,10,5,4)$ in place of $(99,14,1,2)$ are proved false, using the triangular graph $T(7)$ and its automorphism of order 7. The tool `lake comparator` checks that the proofs prove exactly these statements, that they use only the axioms `propext`, `Classical.choice` and `Quot.sound`, and that Lean's kernel accepts them. With the option `-paranoid`, five further checkers bundled with Lean (`leanchecker-paranoid`, `lean4lean`, `nanoda`, `con-leche` and `con-ron`), which re-check every declaration down to Lean's core, accept them as well.

The formal proof follows Sections 3 to 5: Lemma 3.1, the relabelling of the 98 non-fixed vertices, the identities of Proposition 4.1 and Lemma 5.1. For the zero diagonal it uses the

traces of the projections $E_3 = \frac{1}{7}(A + 4I) - \frac{2}{77}J$ and $E_3\Pi_F$, where Π_F averages over the orbits, which give $7d = \operatorname{tr} C + 42$ directly, together with the bound $C_{ii} \le 3$. For Sections 7 and 8 it does not reproduce the hand computations. It uses instead the rank-4 condition in the form $D\,G = G_{\cdot,S}\operatorname{adj}(G_S)\,G_{S,\cdot}$ for explicit four-element sets of indices S, explicit adjugates and $D = \det G_S > 0$. The resulting finite statements, which concern 29 labelled blocks on $\mathcal{A} \cup \mathcal{B}$, are checked by Lean's kernel. The formal proof thus confirms the conclusions of Theorem 7.6 and of Section 8 independently of the computations by hand given in the text.

Use of AI systems. This work was carried out with substantial assistance from AI systems. ChatGPT (OpenAI) produced the first candidate argument, which contains the strategy of the proof: the zero-diagonal step, the rank-4 Gram compression and the case analysis on the LL/RR block. Claude (Anthropic) audited that argument, completed it, wrote out every finite step by hand, drafted the manuscript, ran the exact-arithmetic checks and wrote the Lean formalization. The author directed the work and takes full responsibility for its content.

Acknowledgements. We thank the authors of [CW25], whose framework and notation this paper follows.

Appendix A. Tables

Table 1. Values of $\alpha = 12z_b^2$ from Lemma 7.4, for every column $c = (c_1, c_2, c_3)$ of nonnegative integers with the stated sum s. A negative value means that the column cannot occur.

$P = T_{112}$, $s = 2$	011: 64, 101: 61, 110: 61, 200: 40, 020: 28, 002: 28
$P = T_{112}$, $s = 3$	111: 52, 021: 37, 012: 37, 120: 16, 102: 16, 210: 13, 201: 13, 030: −35, 003: −35, 300: −44
$P = T_{113}$, $s = 2$	011: 64, 110: 52, 101: 52, 200: 40, 020: −8, 002: −8
$P = T_{113}$, $s = 3$	111: 61, 021: 37, 012: 37, 210: 13, 201: 13, 120: −11, 102: −11, 300: −35, 030: −107, 003: −107
$P = T_{113}$, $s = 4$	022: 28, 121: 16, 112: 16, 211: 4, 031: −44, 013: −44, 220: −68, 202: −68, 310: −80, 301: −80, 130: −128, 103: −128, 400: −164, 040: −260, 004: −260
$P = T_{122}$, $s = 3$	111: 62, 201: 38, 021: 38, 102: 32, 012: 32, 210: 8, 120: 8, 300: −40, 030: −40, 003: −58
$P = T_{122}$, $s = 4$	112: 32, 211: 14, 121: 14, 202: 8, 022: 8, 301: −34, 103: −34, 031: −34, 013: −34, 220: −64, 310: −88, 130: −88, 400: −160, 040: −160, 004: −160

Here $c_1c_2c_3 : \alpha$ stands for the column (c_1, c_2, c_3) and its value of α.

TABLE 2. The Gram matrices G^- and G^+ of Lemma 8.1, and their ranks.

case	G^-	$\operatorname{rank} G^-$	G^+	$\operatorname{rank} G^+$
1	$2(3I-J)$	2	$10(3I-J)$	2
2	$\begin{pmatrix} 4 & -2 & -2 \\ -2 & 10 & -8 \\ -2 & -8 & 10 \end{pmatrix}$	2	$\begin{pmatrix} 20 & -10 & -10 \\ -10 & 14 & -4 \\ -10 & -4 & 14 \end{pmatrix}$	2
3	$8(3I-J)$	2	$4(3I-J)$	2
4	$8(3I-J)$	2	$\begin{pmatrix} 8 & -4 & -4 \\ -4 & 8 & 8 \\ -4 & 8 & 8 \end{pmatrix}$	2
5	$\begin{pmatrix} 16 & -8 & -8 \\ -8 & 10 & -2 \\ -8 & -2 & 10 \end{pmatrix}$	2	$\begin{pmatrix} 8 & -4 & -4 \\ -4 & 14 & 14 \\ -4 & 14 & 14 \end{pmatrix}$	2
6	$\begin{pmatrix} 16 & -8 & -8 \\ -8 & 4 & 4 \\ -8 & 4 & 4 \end{pmatrix}$	1	$\begin{pmatrix} 8 & -4 & 8 \\ -4 & 20 & -4 \\ 8 & -4 & 20 \end{pmatrix}$	3

In cases 1–5 every matrix has zero row sums or two equal rows, so it is singular, and it has a nonzero 2×2 principal minor; so it has rank 2. In case 6, G^- has proportional rows, and $\det G^+ = 1728 \neq 0$.

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